

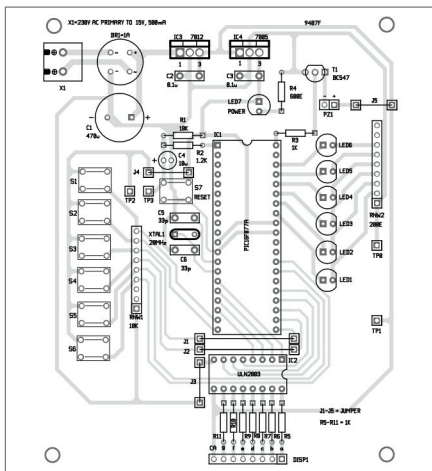


Fig. 5: Component layout for the PCB

(quiz_mm.bas), if everything's fine, you will see 'Compilation Success for Target Device 16F877A (20MHz)' message at the bottom of the screen, as shown in Fig. 3. The hex code generated by Proton IDE is used to burn into the PIC16F877A using PICSTART Plus programmer.

Construction and testing

An actual-size, single-side PCB layout for the circuit is shown in Fig. 4 and

its component layout in Fig. 5. All the components are easily available in electronics components shops. After assembling of the components properly on the PCB, power supply connections should be checked.

Install the 7-segment display DIS1 suitably so that it is visible to the quiz master and the audience. It requires at least eight wires to connect to the PCB. Connection detail for each segment ('a' through 'g') is provided in Fig. 5 at DIS1. Next, install all six switches (S1 through S6) at different locations where the six contesting teams can reach them. House each switch and its corresponding LED in a transparent enclosure, so that when you press the switch and LED glows, it is visible to the quiz master.

Switch on the power supply to start the quiz. On pressing any switch (S1 through S6), the corresponding number (1 through 6) should appear

PARTS LIST

Semiconductors:

- IC1 - PIC16F877A microcontroller
- IC2 - ULN2003 driver
- IC3 - 7812, 12V regulator
- IC4 - 7805, 5V regulator
- T1 - BC547 npn transistor
- BR1 - 1A bridge rectifier
- DIS1 - 12.7cm (5-inch) 7-segment display (common anode)

LED1-LED6

- 5mm white LED (high intensity)
- 5mm red LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 10-kilo-ohm
- R2 - 1.2-kilo-ohm
- R3, R5-R11 - 1-kilo-ohm
- R4 - 680-ohm
- RNW1 - 10-kilo-ohm resistor network
- RNW2 - 200-ohm resistor network

Capacitors:

- C1 - 470 μ F, 35V electrolytic
- C2, C3 - 0.1 μ F ceramic
- C4 - 10 μ F, 16V electrolytic
- C5, C6 - 33pF ceramic

Miscellaneous:

- X_{osc}1 - 20MHz crystal oscillator
- X1 - 230V AC primary to 15V, 500mA secondary transformer
- S1-S6 - Pushbutton switch
- S7 - Tactile switch
- PZ1 - Piezobuzzer

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

on DIS1 and the buzzer should ring. Simultaneously, the corresponding LED near the switch should glow. The LED and display switch off after six seconds while the buzzer stops ringing after two seconds.

When a question is asked by the quiz master, any team can respond by pressing their switch. The switch first pressed is recognised by the microcontroller. Any other switches pressed thereafter do not get registered. So the team who pressed the button first can be allowed to give the answer.

For any problem in the display you can press reset switch S7 momentarily and check again. For further troubleshooting, you can also verify the voltages at various points given in the test points table. ●

Dr D.K. Kaushik is principal and Ashok Sharma is technical assistant at Manohar Memorial (PG.) College, Fatehabad (Haryana)

Would You Like More Hands-on Circuits?



We Have Thousands!

VISIT TODAY

electronicsforu.com

If it's electronics, it's here